

Co-Packaged Optics for AI Infrastructure · Series E · San Jose, CA · Founded 2015

COMPANY OVERVIEW

Ayar Labs is building optical interconnect technology for next-generation AI infrastructure, replacing copper connections inside AI servers with light-based data movement. As GPU clusters scale, copper is becoming constrained by bandwidth, distance, latency, and power consumption, creating a bottleneck for AI data centers. Ayar’s TeraPHY optical I/O chiplet sits adjacent to GPUs and converts electrical signals into light, powered by its SuperNova external laser source. This enables higher bandwidth per watt, longer-reach connectivity, and more efficient scale-up across thousands of GPUs. The company is manufactured by GlobalFoundries today, with next-generation production planned through TSMC, and recently closed a \$500M Series E at a \$3.75B post-money valuation led by Neuberger Berman, with participation from NVIDIA, AMD, MediaTek, QIA, ARK, and others.

WHY CONSIDER

- **Real Silicon, Not Lab Demos:** Ayar has moved beyond concept, with a UCIe-compatible TeraPHY optical chiplet, a 100+ Tbps optical I/O solution with Alchip, and a Wiwynn rack-scale platform designed to connect 1,024+ AI accelerators. The company also has 200+ patents supporting its technical moat.
- **New Tier-1 Lead + Strategic Validation:** Neuberger Berman led the \$500M Series E as a new investor at a \$3.75B valuation, aligning with ID8’s screen. NVIDIA, AMD, MediaTek, Alchip, QIA, ARK, Insight, and others also participated, validating Ayar across both crossover capital and the AI infrastructure ecosystem.
- **Copper Bottlenecks Are Forcing the Shift:** Copper is becoming a power, bandwidth, and distance bottleneck as AI clusters scale. Recent activity across NVIDIA, Marvell/Celestial AI, and AMD/Enosemi suggests photonics is becoming core AI infrastructure, not a niche hardware category.

COMPETITIVE LANDSCAPE

COMPETITOR	DESCRIPTION	THREAT
Marvell (post-Celestial AI) Public co. / Acquired Feb '26	\$3.25B + \$2.25B earnout for Celestial AI. Bundles Photonic Fabric with custom ASIC into hyperscaler deals.	HIGH
Lightmatter \$4.4B Oct '24 / \$822M raised	Direct peer. T. Rowe Price-led. Passage M1000 (114 Tbps) + L200 CPO. Customers deploying late '25.	HIGH
Broadcom (CPO unit) Public co. / Shipping at scale	Already shipping CPO to Meta with 1M+ link-hours, zero failures (Oct '25). Distribution moat through Tomahawk switch silicon.	HIGH
AMD + early-stage CPO startups Strategic + Series A/B	AMD acquired Enosemi for in-house optical. Nubis (Matrix-backed), Xscape (\$44M w/ NVIDIA+Cisco), nEye Sys. (\$58M w/ CapitalG+NVIDIA). Earlier stage.	MEDIUM

ROUND DETAILS	
Round	Series E — Closed March 3, 2026
Size / Pre / Post	\$500M · \$3.25B pre · \$3.75B post
Price/Share	\$52.2418 · Series E Preferred (1x non-part., sr.)
Lead	Neuberger Berman (Gabe Cahill, MD — observer)
Strategics	NVIDIA · AMD · AMD Ventures · MediaTek · Alchip
New Financial	Insight · ARK · Light Street · QIA · Advent · Sequoia Global Eq. · 1789 Capital
Follow-On	Playground · IAG · Boardman Bay · Realm · Etna · Inertia · VentureTech
Use of Proceeds	High-vol. production · test capacity · Hsinchu (Taiwan) · CPO deployment
Prev. Round	Series D: \$155M · \$1.05B/\$1.20B · Dec '24 · Light Street + Advent co-led Series C: \$130M · \$332M/\$452M · Apr '22 · Boardman Bay led Series B: \$35M · \$135M/\$170M · Nov '20 · BlueSky + Downing Series A: \$23.6M · \$36M/\$60M · Nov '18 · Intel Capital + GF Seed: \$2.5M · \$7.5M/\$10M · May '16 · Founders Fund (fka 'OptiBit')

FOUNDING TEAM

- **Mark Wade, Ph.D** Co-Founder & CEO: Photonics pioneer. Led team that designed optics in the world's FIRST processor to communicate using light (MIT/UC Berkeley, 2010-15). Ph.D EE Univ. of Colorado. CEO since Dec '23.
- **Vladimir Stojanovic** Co-Founder & CTO: UC Berkeley professor; technical co-founder. Architected TeraPHY + SuperNova. Targets '10K GPU dies in scale-up at ~100kW rack power.'
- **Board & C-Suite Adds (Mar '26):** Pat Gelsinger (ex-Intel CEO) joined Board alongside Series E. NEW: Sankara Venkateswaran (VP Eng) + Vivek Gupta (CSO, hyperscaler GTM).

DEAL SCORING

AI Fundamentals	4 / 4
Deal Dynamics	4 / 4
Return Profile	3 / 4
Fundamentals	2 / 4
Terms	3 / 4
Overall	3.2 / 4.0

FINANCIAL SNAPSHOT

METRIC	Series C ('22)	Series D ('24)	Series E ('26)
Round Size	\$130M (+ \$10M debt)	\$155M primary	\$500M primary [1]
Pre-Money	\$332M	\$1.05B	\$3.25B [1]
Post-Money	\$452M	\$1.20B	\$3.75B [1]
Price/Share	\$13.81	\$24.47	\$52.24 [1]
Step-Up vs Prev.	~2.7x (B)	~2.7x (C1)	~3.2x (D)
Lead	Boardman Bay	Light St. + Advent	Neuberger Berman [1]
Headcount	90 (Apr '22)	~150 (est.)	196 (Aug '25) [1]
Units Shipped	Demos w/ Intel + DARPA	Prototypes	~15K units (Dec '24)

RISKS & MITIGANTS

Timing — Paying 2026 Price for 2028-2030 Revenue

Yole Group projects mass CPO deployment won't happen until 2028-2030. Ayar is valued at \$3.75B today on revenue still in the low thousands of units. Thesis depends on Wade hitting 100M units/yr by 2028 — a 6,000x ramp in three years. Every quarter of slippage is real value destruction.

Valuation Jumped 3.2x in 15 Months

\$1.2B post in Dec 2024 to \$3.75B post in March 2026 — without a meaningful revenue change. Defensible vs. Lightmatter (\$4.4B) and Celestial AI (\$3.25B + \$2.25B earnouts), but Ayar needs ~\$200M revenue by 2028 to justify entry against Marvell's \$500M FY28 photonics ARR target.

NVIDIA and AMD Could Insource

Both are investors AND building competing in-house capability. NVIDIA's Quantum-X / Spectrum-X switches are NVIDIA-designed, not Ayar-supplied. AMD acquired Enosemi for own optical I/O. Best case: customers buy TeraPHY for accelerator-side, build switches in-house. Worst case: insource everything.

Competitive Landscape is Consolidating

Marvell acquired Celestial AI Feb 2026 — Ayar's closest competitor is now backed by a \$65B public co with built-out hyperscaler sales motion. Marvell guides \$500M photonics ARR by FY28. Lightmatter at \$4.4B has a broader product set. Broadcom shipping CPO to Meta with zero failures.

Capital Intensity Remains High

\$870M raised to date. Deep-tech hardware has high costs — chip tape-outs run \$20-50M each, plus the Taiwan office buildout. Burn \$150-250M/yr now, scaling to \$300M+ by 2027. Series E carries through ~2027; next round needs IPO at \$5B+, strategic deal, or another dilutive round.

Two CEO Transitions Already in the Past

Wright-Gladstein → Wuischpard (2020, clean — she left to start Sphere; remains co-founder/CSO). Wuischpard → Wade (Dec 2023, right before Series D — worth a reference call). Wade has closed Series D and E since at upsized valuations. Pat Gelsinger joining the board in March validates him.

Public comparables for the CPO category · Bull / Base / Bear return scenarios for ID8's Series E entry at \$3.75B post

COMPARABLE CPO / SILICON-PHOTONICS COMPANIES

COMPANY	LATEST VAL	RAISED	LAST ROUND	STAGE	REVENUE	POSITIONING vs AYAR
Ayar Labs	\$3.75B	\$870M	Series E '26	Pre-revenue ramp	~15K units (Dec'24)	Neuberger Berman-led Series E in March. Wade says mid-2026 to mid-2028 is the 'volume production focus window' — targeting 100M+ units/yr by 2028.
Lightmatter	\$4.4B	\$822M	Series D '24	Pre-revenue ramp	Negligible	Closest peer. T. Rowe Price led the Series D in October 2024. Broader product set than Ayar (Passage interposer + Envisie compute chip). Customers expected to deploy late 2025. Not generating meaningful revenue yet either.
Celestial AI (acq.)	\$3.25B + \$2.25B earnout	\$594M	Acquired Feb '26	Acq'd by Marvell	~\$25-50M est.	Marvell announced in December 2025, closed in February. Earnouts: \$500M cumulative revenue by FY29 unlocks first third; \$2B unlocks the full \$2.25B. Sets the strategic exit benchmark for the category.
Broadcom (CPO unit)	\$1.4T public	n/a	Public	Shipping at scale	Embedded in \$5B+ optical biz	Already shipping co-packaged optics to Meta with 1M+ link-hours and zero failures (Oct 2025). Major distribution advantage through Tomahawk switch silicon. The biggest long-term threat if Broadcom pushes harder into accelerator-side CPO.
Marvell (post-Celestial)	\$65B public	n/a	Public	Ramping	Photonic Fabric in \$5B biz	Now bundles Celestial's Photonic Fabric with custom ASIC and Teralynx switches into hyperscaler deals. Told Wall Street to expect \$500M photonics ARR by FY28 and \$1B by FY29. That's the public-market bar Ayar is being judged against.
Xscape / Nubis / nEye Sys.	Series A/B funded	\$44M / undisc / \$58M	Series A/B '24-'25	Early R&D	Pre-revenue	Xscape (IAG-led with Cisco + NVIDIA), Nubis (Matrix + MediaTek), nEye (CapitalG-led with NVIDIA + M12 + Micron). Earlier stage, smaller capital. Not near-term threats but worth tracking — they're the next wave.

The read-across: Ayar at \$3.75B sits below Lightmatter (\$4.4B) on a similar pre-revenue-ramp base. It's above Celestial AI's \$3.25B headline acquisition price but below the \$5.5B figure once earnouts are included. Marvell's guidance to Wall Street — \$500M ARR in photonics by FY28 — tells us the public market accepts roughly 7x forward revenue for shipped-and-scaling CPO. To defend \$3.75B against that comp, Ayar needs to clear \$200M+ in revenue by 2028.

BULL / BASE / BEAR SCENARIOS — ID8 ENTRY AT \$3.75B

BULL	4 – 7× over 3-4 yrs	BASE	1.5 – 2.5× over 4-5 yrs	BEAR	0.5 – 1.0× (flat to loss)
Mass CPO deployment hits 2028 as Wade plans. NVIDIA and AMD buy TeraPHY for the accelerator side, and the hyperscalers (Microsoft, Meta, Amazon, Google) come in for incremental volume. Ayar captures ~25% of the \$5-20B optical I/O market = \$1-2B in revenue by 2029-2030, IPO at \$15-25B. TRIGGERS / WATCH-FOR: What to watch for: TSMC's COUPE platform qualifies on schedule · Top-3 hyperscaler issues a binding PO by end of 2026 · AMD's MI400 ships with TeraPHY inside · Marvell hits its FY28 \$500M photonics ARR target		Mass deployment slips one year to 2029-2030. Competition is tougher than expected as Marvell and Broadcom bundle photonics into bigger deals. Ayar captures ~15% share = \$400-700M in revenue by 2030, IPO or strategic exit at \$6-10B. Crossover investors (Neuberger, Insight, ARK, QIA) have the hold period to wait it out. TRIGGERS / WATCH-FOR: What to watch for: COUPE slips 6-12 months · NVIDIA insources the accelerator side · Hyperscalers default to Broadcom bundles · Marvell ramps Photonic Fabric faster than Ayar ramps TeraPHY		Mass deployment slips past 2030. Marvell + Broadcom bundling wins the category. Ayar's share is 5-10% = \$100-300M in revenue, leading to a down round or a strategic exit at \$2-4B (the buyer set narrowed when Marvell took Celestial off the board). TRIGGERS / WATCH-FOR: What to watch for: COUPE qualification fails or slips >12 months · NVIDIA insources both accelerator AND switch-side CPO · Pluggable optics advance faster than expected · Capital markets close before the 2027 round	

Research date: May 2026 · PitchBook (PBID 113119-75) Priority 1 · Live web search verified May '26 · Numbers in slides ([1] [2] [3]...) reference rows below

#	SOURCE / PUBLICATION	WHAT WAS FOUND	URL / CITATION
1	PitchBook (PBID 113119-75)	Founded 2015 (former name OptiBit); legal entity Ayar Labs, Inc.; HQ 695 River Oaks Parkway, San Jose CA 95134. Series E closed Mar 3 '26 at \$3.25B pre / \$3.75B post on \$500M raise. Investor ownership 57.49% post-E. 196 employees Aug '25. Cumulative raised \$870M per company PR (\$874M per PitchBook).	https://my.pitchbook.com/profile/113119-75/company/profile
2	PitchBook deal record — Series E (deal 323581-06T)	\$500M deal Mar 3 '26; price/share \$52.2418; Series E Preferred (1x non-participating, weighted-avg anti-dilution, senior liquidation). 25 total investors (17 new / 8 follow-on). 13.16% acquired. Counsel: Latham (Ayar), Cooley (Neuberger), Willkie (Insight), Weil (Advent).	https://my.pitchbook.com/profile/113119-75/company/deals#deal-323581-06T
3	Ayar Labs press release + Optics & Photonics News + BusinessWire (Mar 3 '26)	Series E led by Neuberger Berman; new investors Alchip, ARK Invest, Insight, MediaTek, QIA, Sequoia Global Eq., 1789 Capital. Use of proceeds: high-volume production, test capacity, Hsinchu (Taiwan) office, CPO deployment. Neuberger MD Gabe Cahill: 'execution against key customer milestones driving strong market conviction.'	https://ayarlabs.com/news/ayar-labs-closes-500m-series-e-accelerates-volume-production-of-co-packaged-optics/
4	The Register + Tom's Hardware + SiliconANGLE — Product & Tech (Mar '26)	TeraPHY chiplets manufactured by GlobalFoundries (current) and TSMC COUPE (next-gen). SuperNova external multi-wavelength light source. 100+ Tbps demo'd with Alchip on TSMC COUPE at SC25 (Nov '25). Targets 10K+ GPU dies in scale-up domain at ~100kW rack power per CTO Stojanovic exclusive EI Reg interview.	https://www.theregister.com/2026/03/03/ayar_labs_500m/
5	Ayar Labs + Wiwynn rack-scale partnership (Mar 11 '26)	Joint AI CPO solution showcased at OFC 2026 (Mar 15-19, LA). 1,024-GPU rack-scale reference platform at 100-200kW vs NVIDIA NVL72 at 600kW. Wiwynn has shipped servers to 750+ data centers worldwide. Liquid-cooled HVDC architecture. CEO Wade: 'AI infrastructure is outgrowing copper.'	https://www.theregister.com/2026/03/11/ayar_labs_wiwynn_photonics/
6	Pat Gelsinger Board Announcement (X / Mar '26) + leadership additions	Pat Gelsinger (former Intel CEO) joined Ayar Labs Board March 2026 alongside Series E. Pioneered silicon photonics research at Intel 20+ years ago. NEW hires: Sankara Venkateswaran (VP Engineering, network acceleration silicon) + Vivek Gupta (CSO, leads hyperscaler collaborations).	https://x.com/PGelsinger/status/1990824961519595932
7	PitchBook deals — Ayar Series A through D + GUC partnership	Series D: \$155M / \$1.2B Dec '24 (Light Street + Advent co-led). Series C1: \$25M / \$500M Apr '23 (IAG + Capital TEN). Series C: \$130M / \$452M Apr '22 (Boardman Bay led \$120M equity + \$10M debt). Series B: \$35M / \$170M Nov '20. Series A: \$23.6M / \$60M Nov '18 (Intel Capital + GF). GUC partnership Nov '25: 200+ Tbps aggregate bandwidth.	https://my.pitchbook.com/profile/113119-75/company/deals
8	Marvell + Celestial AI acquisition (Marvell PR / Feb '26) — comp	Marvell completed \$3.25B acquisition of Celestial AI Feb '26 (announced Dec '25); up to \$5.5B with earnout. Earnout milestones: \$500M cumulative rev FY29 = 1/3 unlock, \$2B = full. Marvell guides \$500M ARR (FY28), \$1B (FY29) photonics. Sets strategic exit + public-market comp benchmark for Ayar.	https://investor.marvell.com/news-events/press-releases/detail/1000/marvell-to-acquire-celestial-ai
9	Lightmatter Series D + Contrary Research — competitor comp	Lightmatter closed \$400M Series D Oct '24 at \$4.4B post, T. Rowe Price-led. Total raised \$822M (PitchBook) / \$850M (Lightmatter). Passage M1000 (114 Tbps) + L200/L200x CPO (Apr '25, 200+ Tbps). Customers expected to deploy late '25 per CEO Klinger. 60+ patent families (Mar '25). 315 employees Mar '26.	https://research.contrary.com/company/lightmatter
10	SemiAnalysis CPO Book + EDN + Yole Group — Market & Tech	TeraPHY uses monolithic CMOS+SiPho on GlobalFoundries Fotonix; pivoting to TSMC COUPE heterogeneous integration for next-gen. Microring resonator architecture. 200+ patents. Yole projects mass CPO deployment 2028-2030. Wade public framing: mid-26 to mid-28 'two-year volume production focus window' → 100M+ units/yr by 2028.	https://newsletter.semianalysis.com/p/co-packaged-optics-cpo-book-scaling